Section 4.4: The Normal Distribution

Finite Mathematics MTH 107 Fall 2014

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Mars Hill University

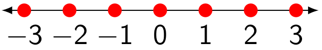
Wednesday 5th November, 2014

Discrete vs Continuous Variables

Discrete

ex. Number of students in class

ex. Integers:



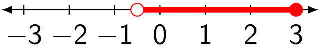
ex. Votes for a canidate

ex. Money: 1, .25, .10, .05, .01

Continuous

ex. The real numbers

ex. $-1/2 < x \leq 3$



ex. height, weight, time, etc.

The heights of $10,000$ randomly selected men.

Histogram

Heights of Men

The heights of 10,000 randomly selected men.

- $n = 10,000$

Histogram

Heights of Men

The heights of 10,000 randomly selected men.

- $n = 10,000$
- $\bar{x} = 70$ inches

Histogram

Heights of Men

The heights of 10,000 randomly selected men.

- $n = 10,000$
- $\bar{x} = 70$ inches
- $s = 3$ inches

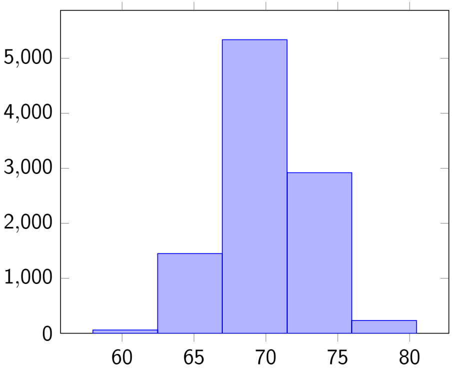
Histogram

Heights of Men

The heights of 10,000 randomly selected men.

- $n = 10,000$
- $\bar{x} = 70$ inches
- $s = 3$ inches

Histogram



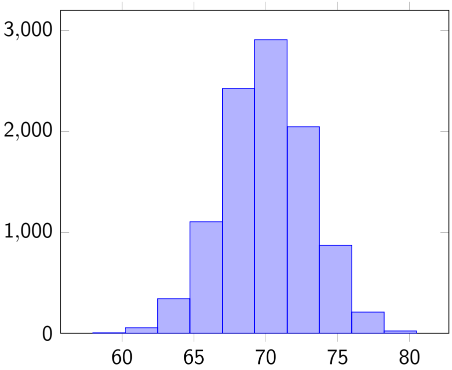
- 5 groups

Heights of Men

The heights of 10,000 randomly selected men.

- $n = 10,000$
- $\bar{x} = 70$ inches
- $s = 3$ inches

Histogram

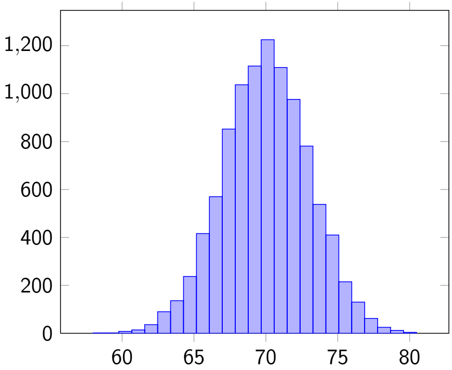


- 10 groups

The heights of $10,000$ randomly selected men.

- $n = 10,000$
- $\bar{x} = 70$ inches
- $s = 3$ inches

Histogram

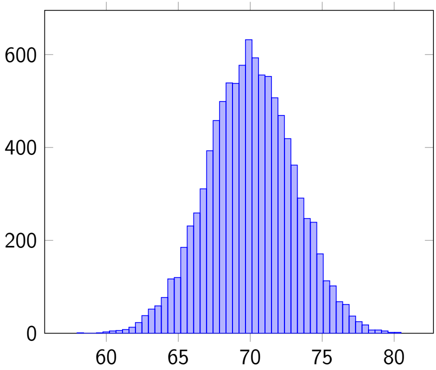


- 25 groups

The heights of 10,000 randomly selected men.

- $n = 10,000$
- $\bar{x} = 70$ inches
- $s = 3$ inches

Histogram

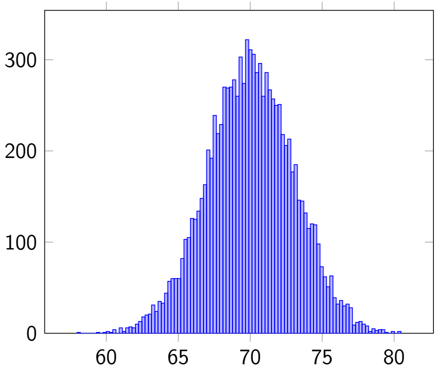


- 50 groups

The heights of 10,000 randomly selected men.

- $n = 10,000$
- $\bar{x} = 70$ inches
- $s = 3$ inches

Histogram



- 100 groups

Sums of 10,000 randomly generated numbers.

Histogram

Sums of 10,000 randomly generated numbers.

- $n = 10,000$

Histogram

Standard Normal Distribution

Sums of 10,000 randomly generated numbers.

- $n = 10,000$
- $\bar{x} = 0$

Histogram

Standard Normal Distribution

Sums of 10,000 randomly generated numbers.

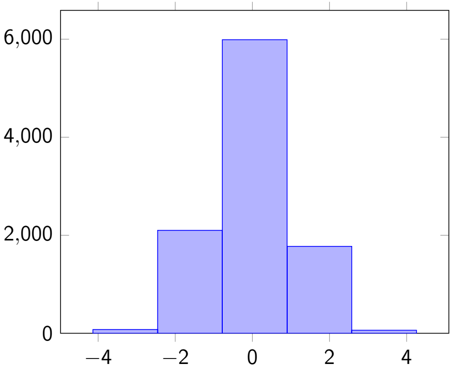
- $n = 10,000$
- $\bar{x} = 0$
- $s = 1$

Histogram

Sums of 10,000 randomly generated numbers.

- $n = 10,000$
- $\bar{x} = 0$
- $s = 1$

Histogram

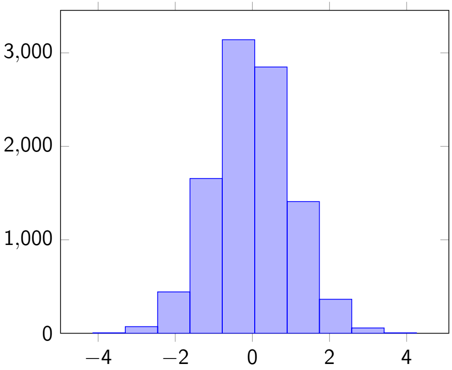


- 5 groups

Sums of 10,000 randomly generated numbers.

- $n = 10,000$
- $\bar{x} = 0$
- $s = 1$

Histogram



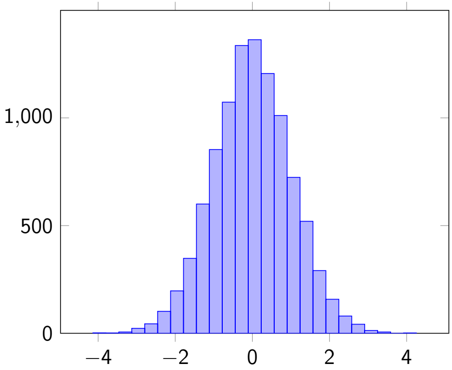
- 10 groups

Standard Normal Distribution

Sums of 10,000 randomly generated numbers.

- $n = 10,000$
- $\bar{x} = 0$
- $s = 1$

Histogram

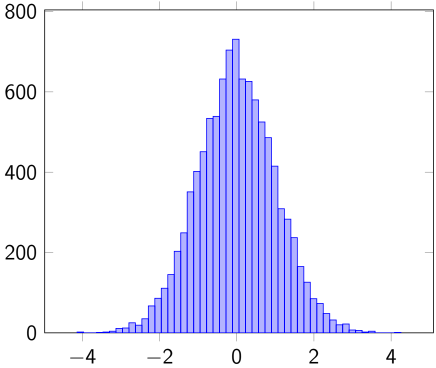


- 25 groups

Sums of 10,000 randomly generated numbers.

- $n = 10,000$
- $\bar{x} = 0$
- $s = 1$

Histogram



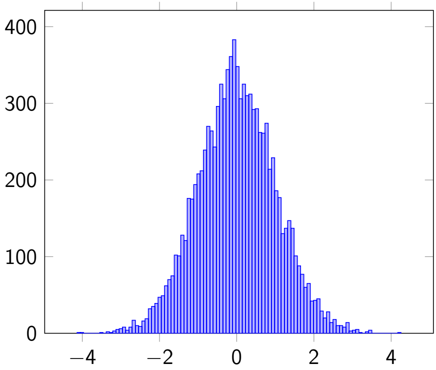
- 50 groups

Standard Normal Distribution

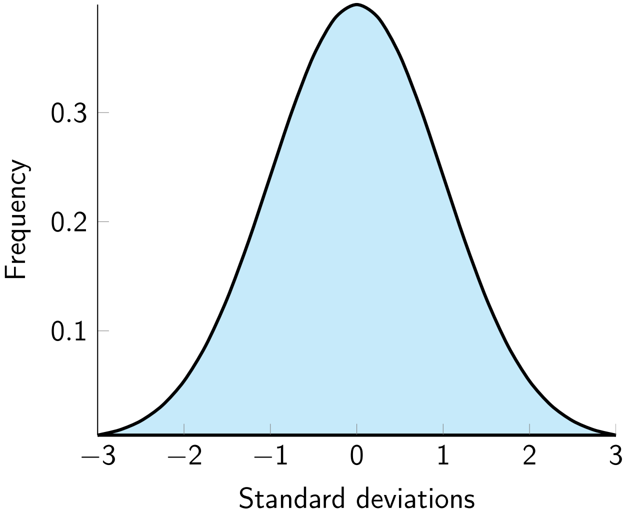
Sums of 10,000 randomly generated numbers.

- $n = 10,000$
- $\bar{x} = 0$
- $s = 1$

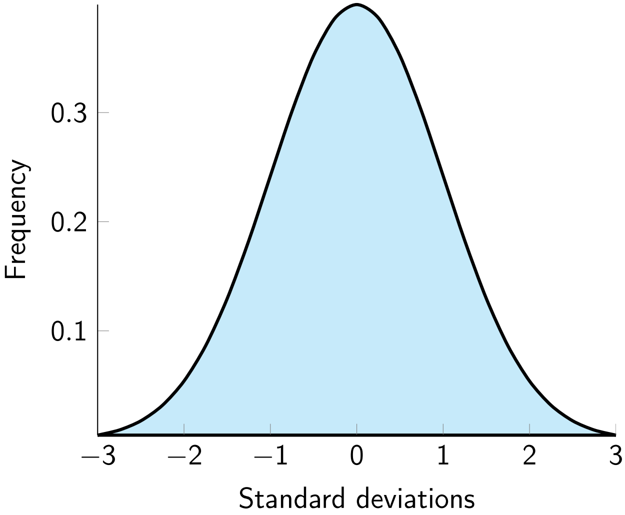
Histogram



- 100 groups

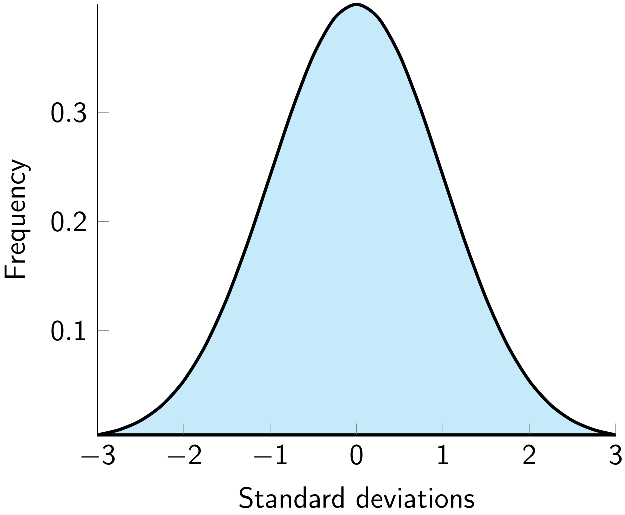

0.3
0.2
0.1
Frequency
−3
−2
−1
0
1
2
3
Standard deviations

Normal Distribution



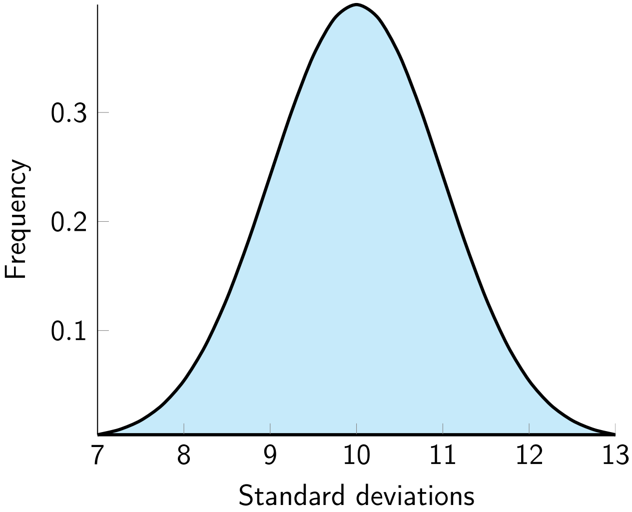
$\bar{x} = 0$

Normal Distribution

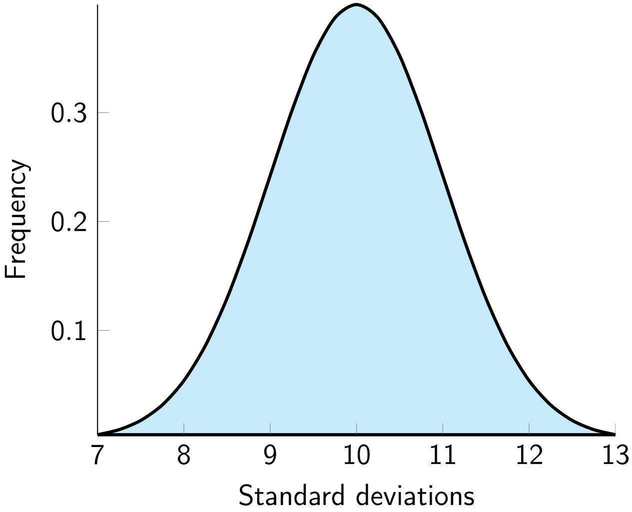


$\bar{x} = 0$

$s = 1$

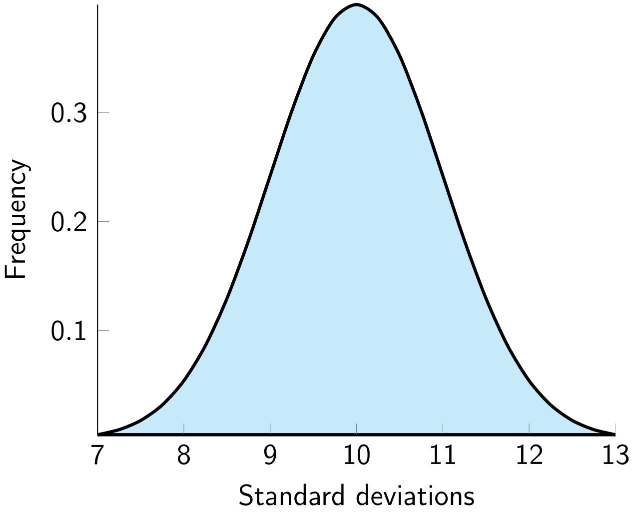

0.3
0.2
0.1
Frequency
7
8
9
10
11
12
13
Standard deviations

Normal Distribution



$\bar{x} = 10$

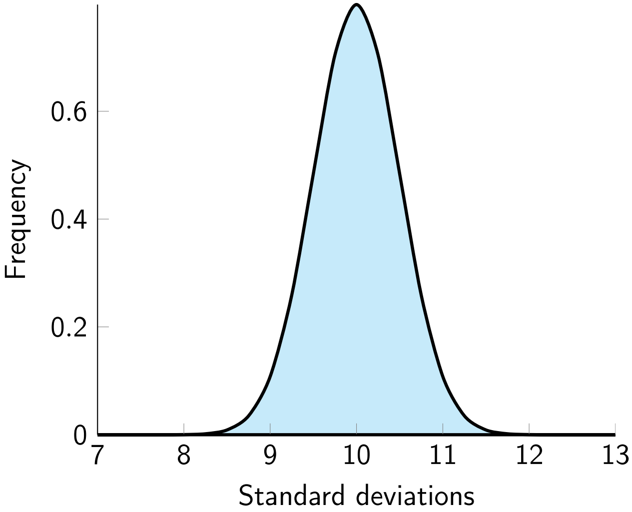
Normal Distribution



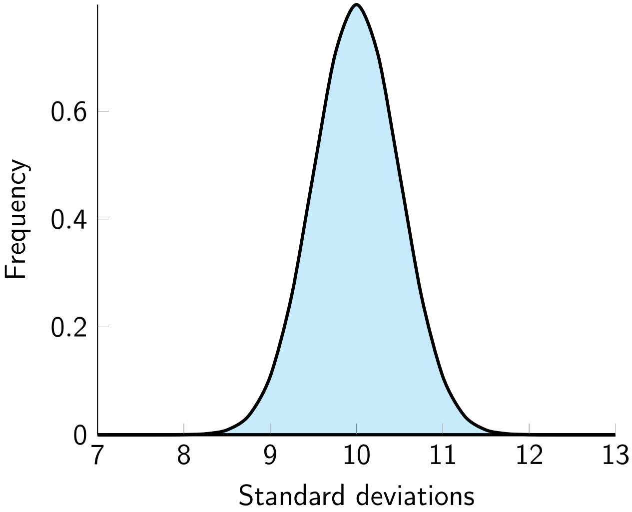
$\bar{x} = 10$

$s = 1$

Normal Distribution

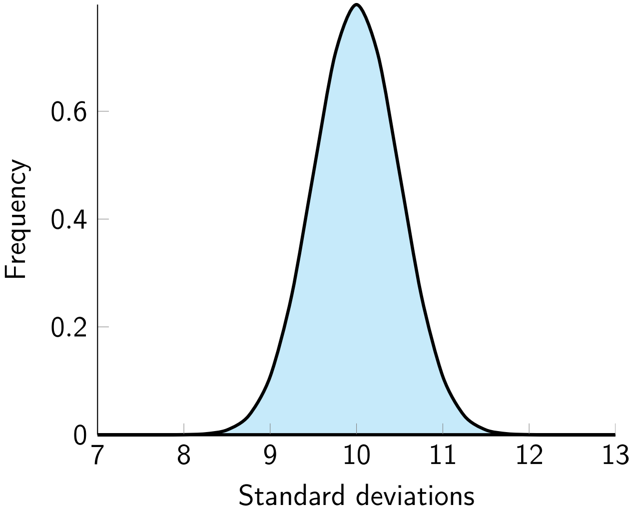


Normal Distribution



$\bar{x} = 10$

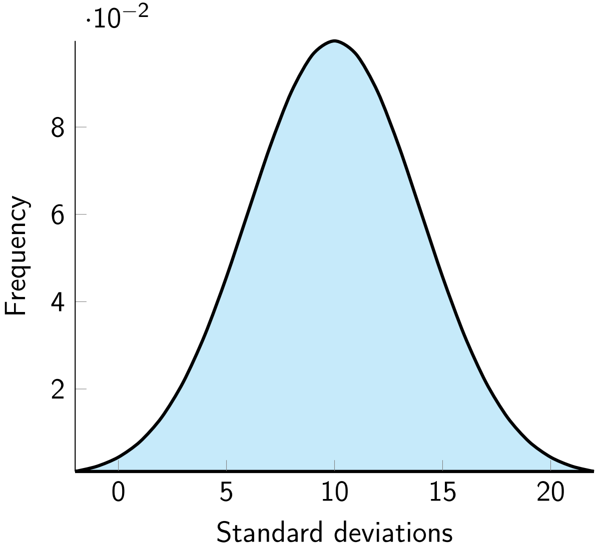
Normal Distribution

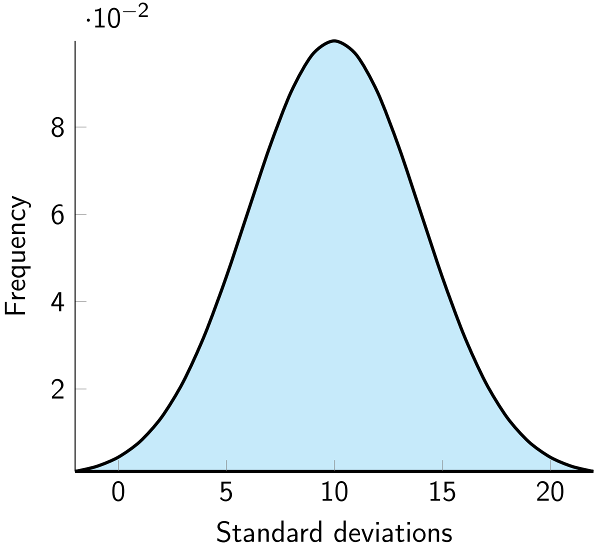


$\bar{x} = 10$

$s = .5$

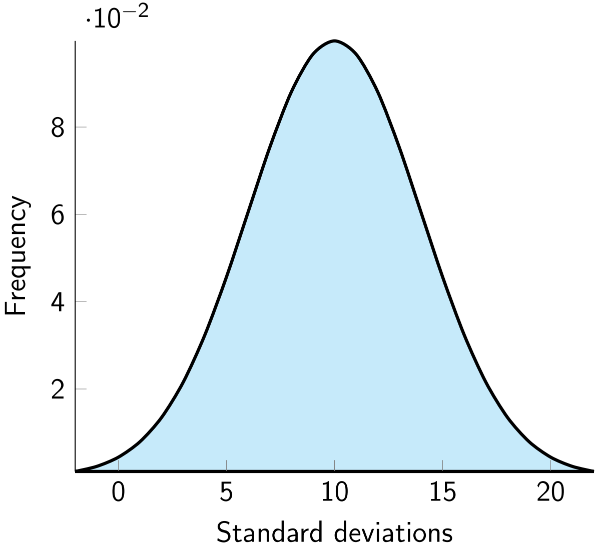
Normal Distribution





$\bar{x} = 10$

Normal Distribution



$\bar{x} = 10$

$s = 4$

notiation

Sample

Population

Sample

$\bar{x}$: means

Population

Sample

$\bar{x}$: means

s: standard deviation

Population

Sample

$\bar{x}$: means

s: standard deviation

Population

μ: means

Sample

$\bar{x}$: means

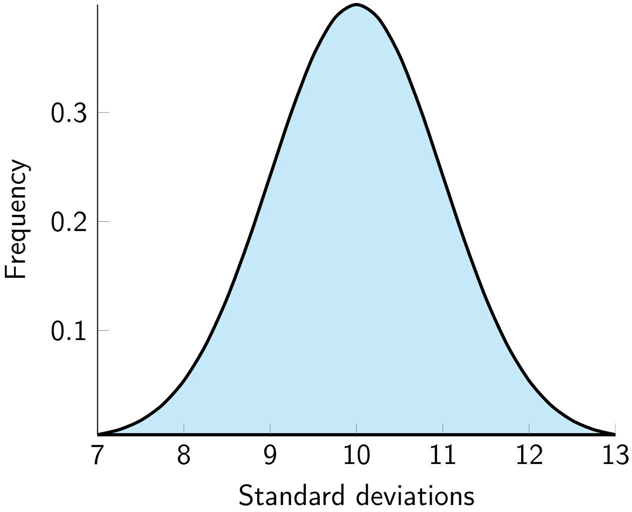
s: standard deviation

Population

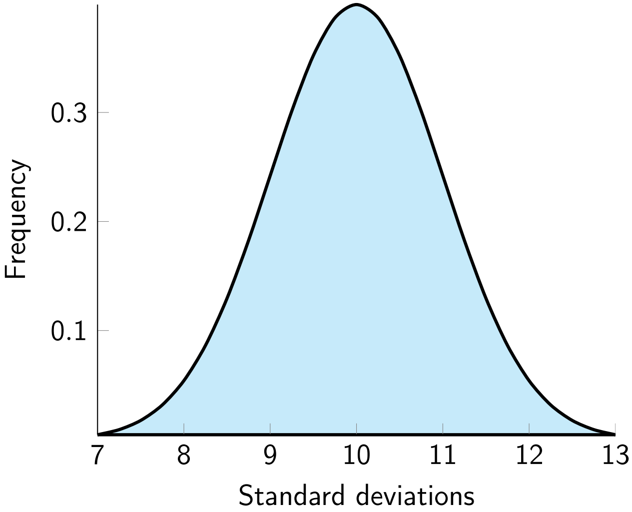
μ: means

σ: standard deviation

Normal Distribution

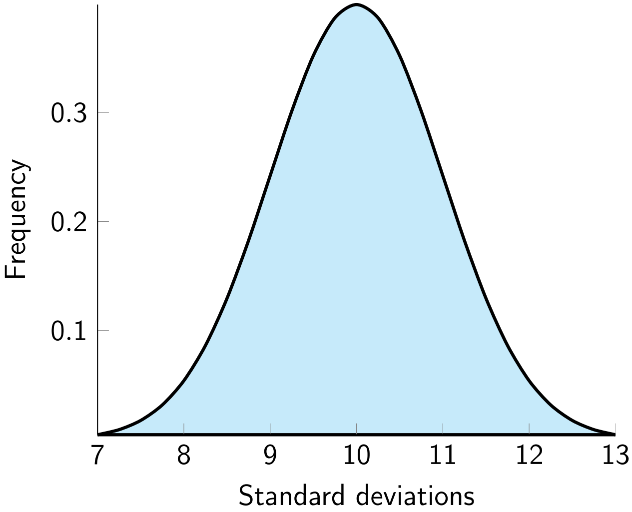


Normal Distribution



$\bar{x} = 10$

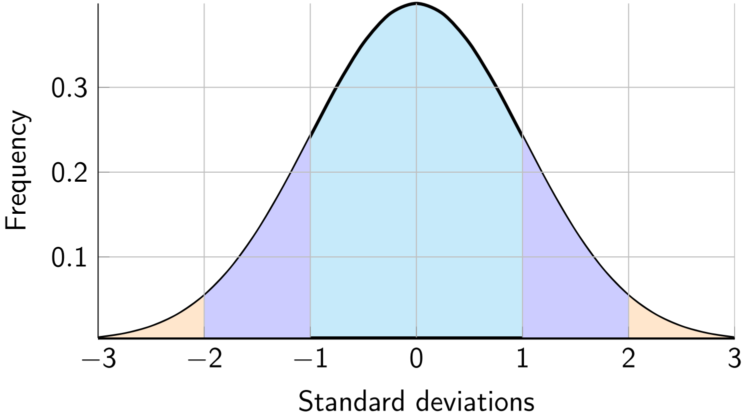
Normal Distribution



$\bar{x} = 10$

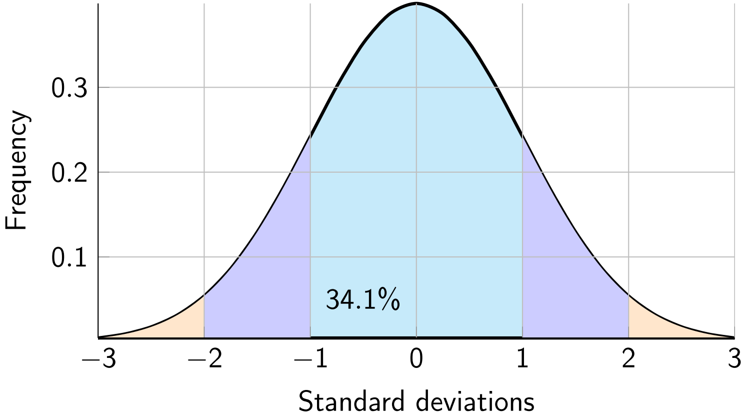
$s = 1$

Normal Distribution



$\mu = 0$ and $\sigma = 1$ (σ for the std dev. of the *population*.)

Normal Distribution



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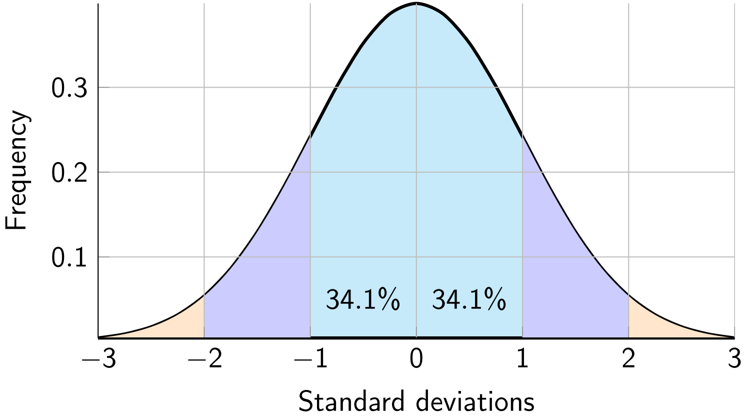
Normal Distribution



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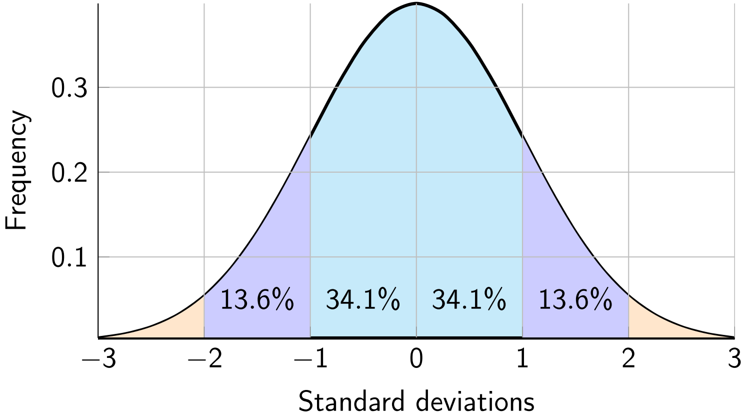
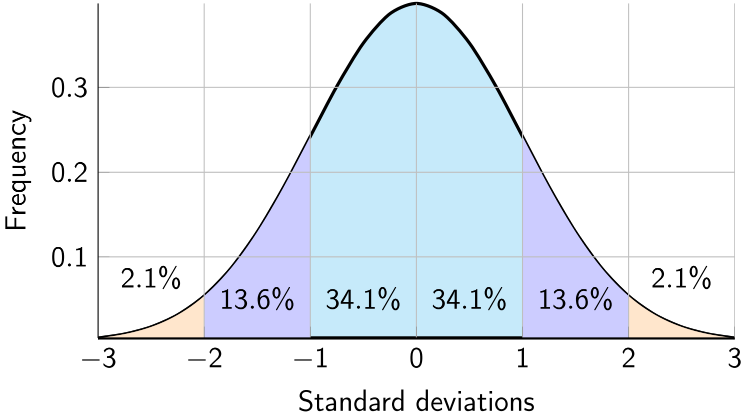
Normal Distribution



$\mu = 0$ and $\sigma = 1$ (σ for the std dev. of the *population*.)

Normal Distribution



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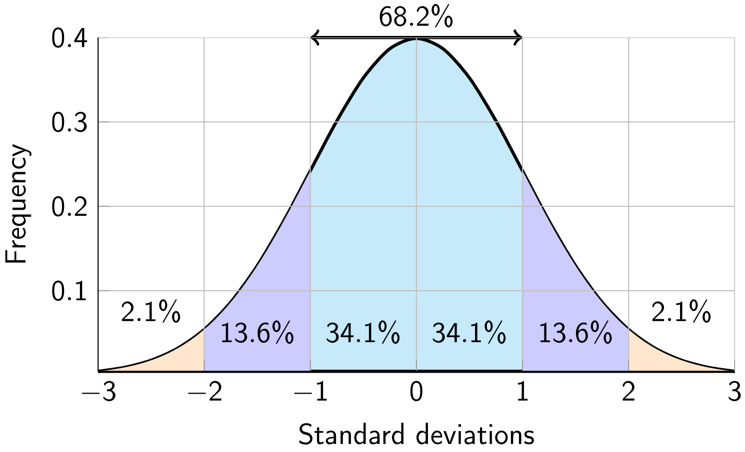
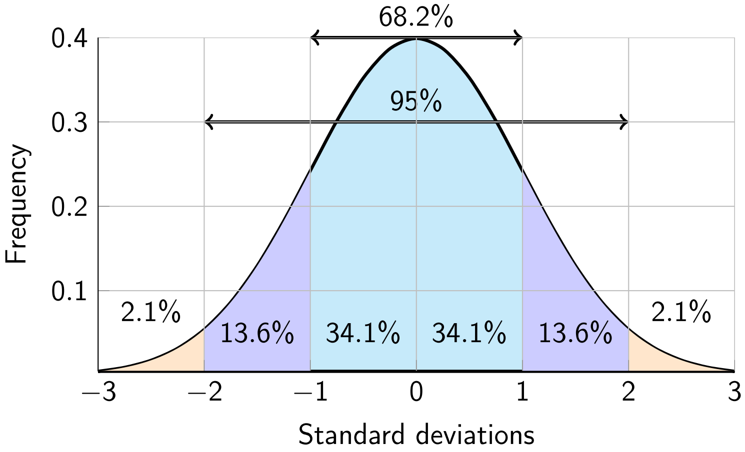
Normal Distribution



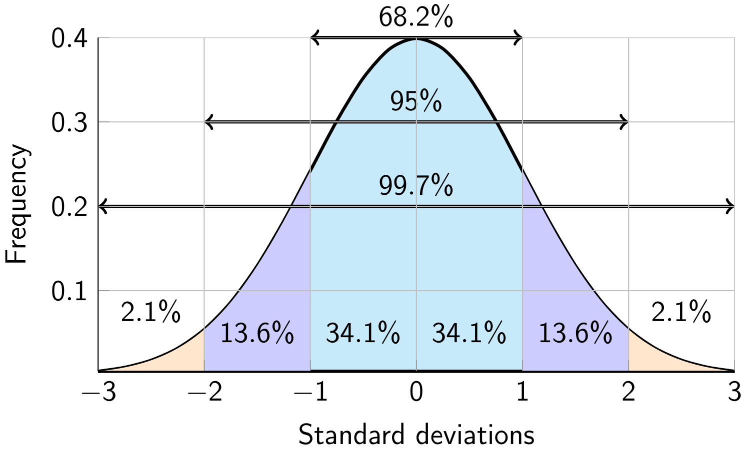
$\mu = 0$ and $\sigma = 1$ (σ for the std dev. of the *population*.)

Normal Distribution



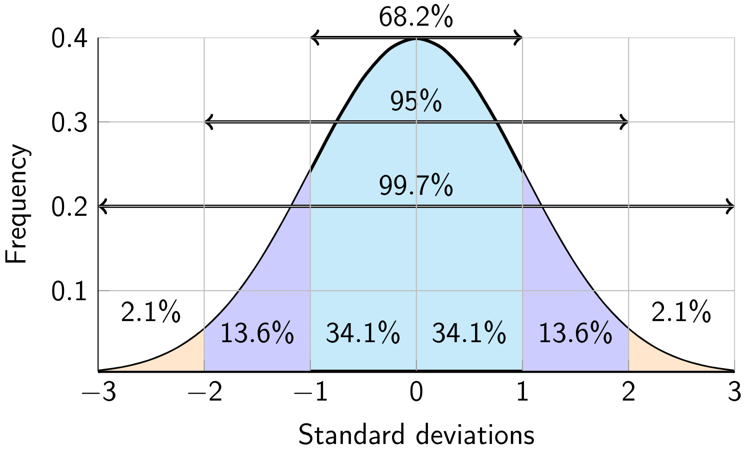
$\mu = 0$ and $\sigma = 1$ (σ for the std dev. of the *population*.)

Normal Distribution



$\mu = 0$ and $\sigma = 1$ (σ for the std dev. of the *population*.)

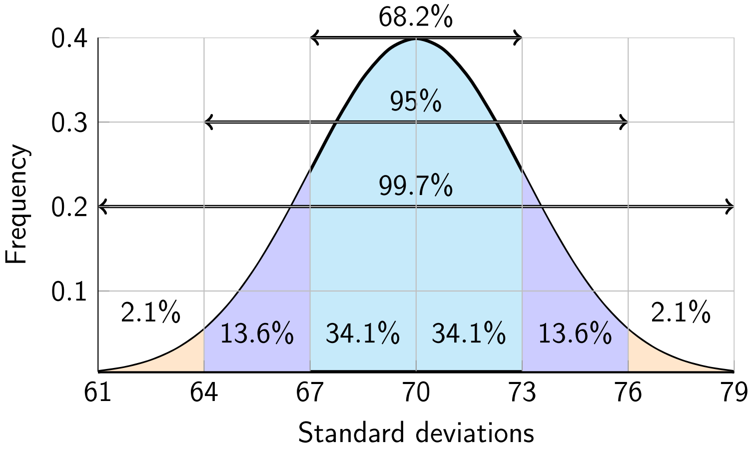
Normal Distribution



$\mu = 0$ and $\sigma = 1$ (σ for the std dev. of the *population.*)

- *Almost* all data is within 3 standard deviations of the mean (.3% on outide or 0.15% on each tail)

Normal Distribution



$\mu = 70$ and $\sigma = 3$

- *Almost* all data is within 3 standard deviations of the mean

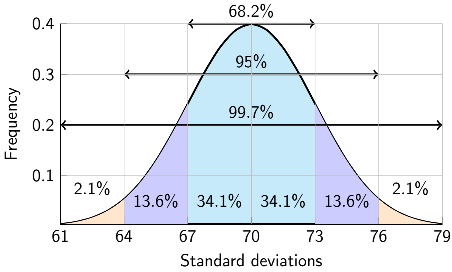
Example 4.4.1.

The mean height of males is 70 inches with a standard deviation of 3 inches.

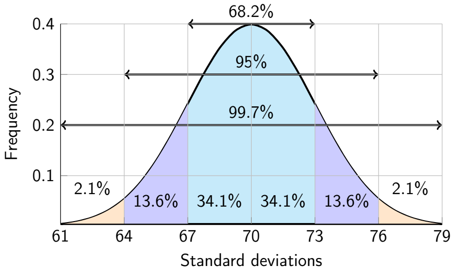
a. What percentage of men are between 67 and 73 inches?

b. What percenatge of men are taller than 67 inches?

c. What percentage of men are at least 76 inches?

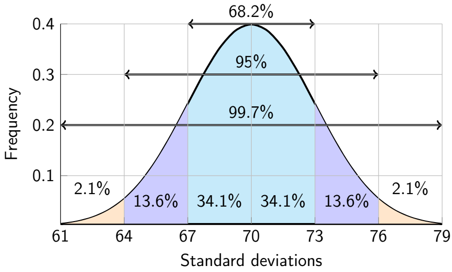


Normal Distribution



a. 68.2%

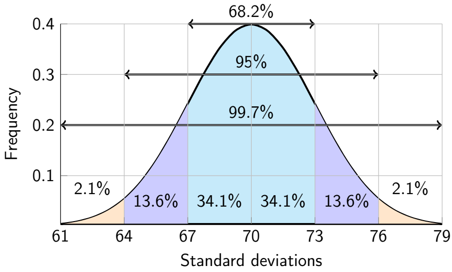
Normal Distribution



a. 68.2%

b. $68.2\% + 13.6\% + 2.1\% + .15\% = 84.05\%$

Normal Distribution



a. 68.2%

b. $68.2\% + 13.6\% + 2.1\% + .15\% = 84.05\%$

c. $13.6\% + 2.1\% + .15\% = 15.85\%$

Example 4.4.2.

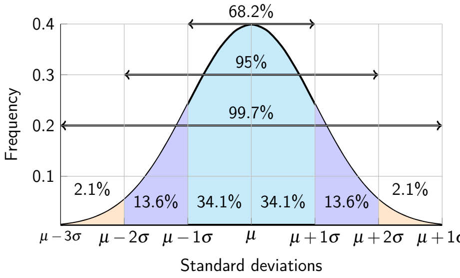
The mean height of females is 64 inches with a standard deviation of 3 inches.

a. What percentage of women are between 61 and 67 inches?

b. What percenatge of women are taller than 70 inches?

c. Which is taller, a 70 inch woman or a 73 inch man?

d. Which is more rare, a 70 inch woman or a 73 inch man?



Example 4.4.2.

The mean height of females is 64 inches with a standard deviation of 3 inches.

a. What percentage of women are between 61 and 67 inches?
b. What percenatge of women are taller than 70 inches?
c. Which is taller, a 70 inch woman or a 73 inch man?
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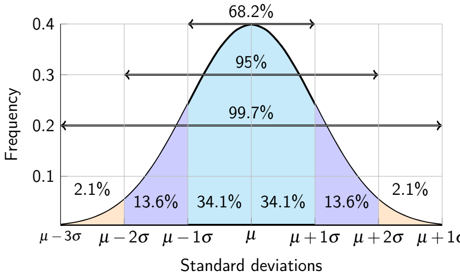

a. 68.2%

Normal Distribution

Example 4.4.2.

The mean height of females is 64 inches with a standard deviation of 3 inches.

a. What percentage of women are between 61 and 67 inches?
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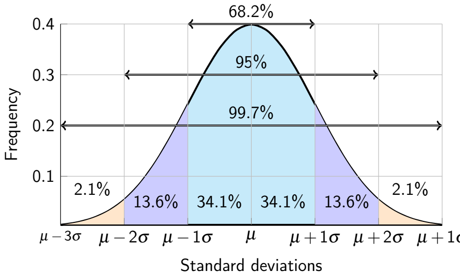

a. 68.2%
b. $2.1\% + .15\% = 2.25\%$

Example 4.4.2.

The mean height of females is 64 inches with a standard deviation of 3 inches.

a. What percentage of women are between 61 and 67 inches?

b. What percenatge of women are taller than 70 inches?

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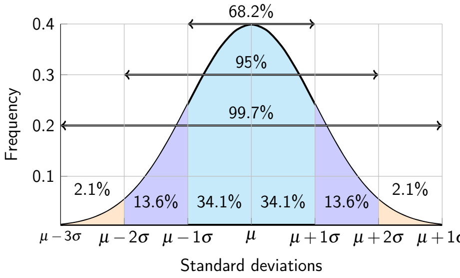


a. 68.2%

b. $2.1\% + .15\% = 2.25\%$

c. The man

Normal Distribution

Example 4.4.2.

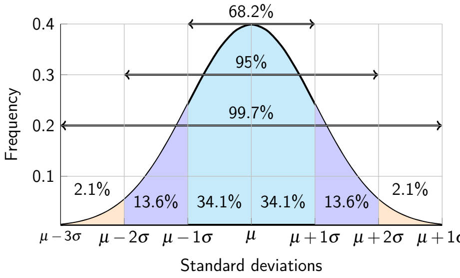
The mean height of females is 64 inches with a standard deviation of 3 inches.

a. What percentage of women are between 61 and 67 inches?

b. What percenatge of women are taller than 70 inches?

c. Which is taller, a 70 inch woman or a 73 inch man?

d. Which is more rare, a 70 inch woman or a 73 inch man?



a. 68.2%

b. $2.1\% + .15\% = 2.25\%$

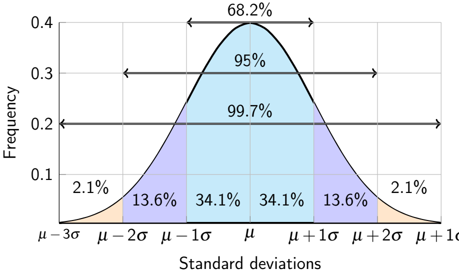
c. The man

d. but a 70 inch woman is more rare. Relatively, she is a "taller" woman.

Example 4.4.3.

The mean height of females is 64 inches with a standard deviation of 3 inches.

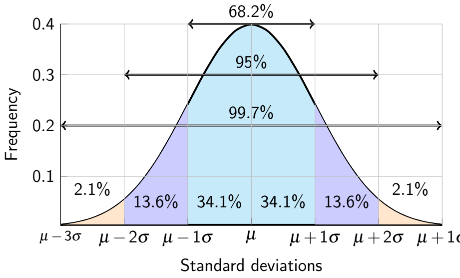
a. Which is more rare, a 65 inch woman or a 71 inch man?



Example 4.4.3.

The mean height of females is 64 inches with a standard deviation of 3 inches.

a. Which is more rare, a 65 inch woman or a 71 inch man?



a. The woman